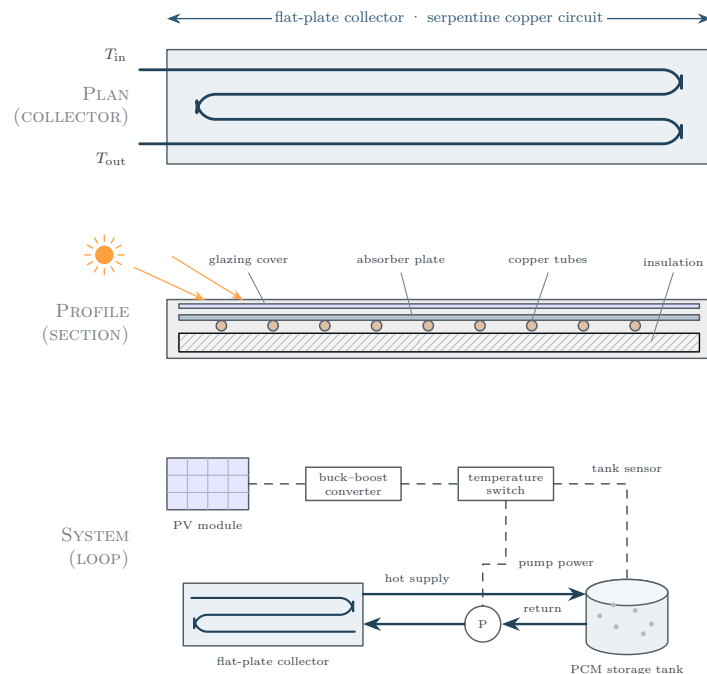


Solar Water Heating System

with PCM-Based Thermal Energy Storage

Design and Development of a Solar Water Heating System with PCM-Based Thermal Energy Storage

REE441 · Senior Design Project II · Full-Term Project Report



General arrangement (not to scale). The serpentine flat-plate collector, PCM storage tank, and PV-powered circulation loop.

Authors: Salman Khalid Alshammari (201904914)
Hakeem Faisal Almohawes (201903115)
Hassan Bassam Qarani (201907603)
Advisor: Prof. Dr. Tawfik Guesmi
Course: REE441 — Senior Design Project II
Document: Full-Term Project Report · 2025

Acknowledgments

We would like to express our sincere gratitude to our advisor, Prof. Tawfik Guesmi, for his continuous guidance, support, and encouragement throughout the course of this senior design project. His expertise, insightful feedback, and dedication were instrumental in shaping the direction of our work and overcoming the various challenges we encountered. We are truly grateful for the opportunity to learn under his supervision and for the valuable experience this project has provided.

Contents

Acknowledgments	i
Table of Figures	iv
Abstract	v
1 Project Definition	1
2 Objectives	2
3 Literature Review	4
3.1 Flat Plate Solar Collector Design	4
3.2 Phase Change Materials (PCMs)	4
3.3 Solar Module	5
3.3.1 Monocrystalline Modules	6
3.3.2 Polycrystalline Modules	6
3.4 DC–DC Converters	7
3.4.1 Buck Converters	7
3.4.2 Boost Converters	7
3.4.3 Buck–Boost Converters	7
3.4.4 Temperature Switch	8
3.4.5 DC Brushless Water Pump	9
4 Strategy and Plan of Action	10
4.1 Phase 1: Design and Component Selection	10
4.2 Phase 2: Computational Analysis (CFD Simulations)	10
4.3 Phase 3: Prototype Development and Experimental Testing	10
4.4 Phase 4: Performance Optimization and Final Validation	11
5 Risk Assessment and Mitigation Strategies	11
5.1 Material Failure	11
5.2 Inaccurate Computational Fluid Dynamics (CFD) Results	11
5.3 Heat Loss	11
5.4 Pressure Drop and Flow Instabilities	12
5.5 PCM Performance Degradation	12
5.6 Data Acquisition and Sensor Failure	12
5.7 Environmental and Weather Conditions	12
5.8 Cost and Manufacturing Constraints	13
6 Key Milestones and Timeline	13
7 Design Methodology and Equations	15
7.1 Material Selection and Justification	15
7.1.1 Copper Pipes	15
7.1.2 Water Tank	15
7.1.3 Phase Change Materials (PCMs)	15
7.2 Thermal Efficiency and Heat Transfer Calculations	16
7.3 Experimental Tools	17
7.3.1 Solar Irradiance Meter	17
7.3.2 Thermometer	17

8	Testing and Results	18
8.1	Flat Plate	18
8.2	PCM Storage Tank	19
8.3	Overall System Performance	20
9	Conclusion	22
A	Appendix A: SDP Relevant Courses, Standards and Constraints	23
	References	24

Table of Figures

1	The actual project designed.	1
2	Schematic diagram of flat plate serpentine design.	2
3	PCM storage tank.	2
4	Flat plate collector components.	4
5	Working of phase change materials.	5
6	The monocrystalline module used in this project.	6
7	Monocrystalline and polycrystalline solar modules.	6
8	Shows the buck–boost converter used in this project.	7
9	Temperature switch used in this project.	8
10	DC pump used in this project.	9
11	Paraffin PCM used in this project.	15
12	Solar irradiance meter.	17
13	Digital thermometer.	17
14	T_{in} vs T_{out}	18
15	Q_u vs Q_t	19
16	Solar irradiance.	19
17	PCM tank.	20
18	Overall system.	22

Abstract

This report presents the design, development, and experimental evaluation of a fully renewable solar water heating system that combines a flat-plate solar collector with a phase change material (PCM)-based thermal storage tank. The primary objective of the project was to achieve stable water temperatures in the range of 60–80 °C while enhancing thermal efficiency and extending thermal storage capacity. The flat-plate collector was tested under real environmental conditions, demonstrating a reliable thermal efficiency of 63%. To address the challenge of energy retention during periods of low or no solar radiation, a PCM storage tank was integrated into the system, utilizing the latent heat properties of the material to maintain higher water temperatures. Comparative testing showed that the PCM-enhanced tank retained heat significantly longer than a conventional water tank, maintaining a temperature 9 °C higher after 12 hours of cooling. Experimental testing demonstrated that the system is capable of heating approximately 30 liters of water per day, raising the temperature from an initial 23 °C to 80 °C under typical sunny conditions with an average daily solar radiation of 6.2 kWh/m²/day. The system operates entirely on renewable solar energy, without reliance on the electrical grid.

1. Project Definition

In recent years, global concerns over climate change, energy security, and the depletion of fossil fuel resources have increased the focus on sustainable energy solutions. Among the various renewable energy technologies, solar thermal systems have emerged as one of the most practical and environmentally friendly alternatives for reducing carbon emissions and meeting growing energy demands. Solar thermal systems utilize the sun's abundant energy to produce heat, which can be used for water heating, space heating, and even industrial processes. They offer a clean and cost-effective means to reduce reliance on conventional energy sources, making them an integral part of future energy strategies.

Flat plate solar collectors (FPCs) are among the most commonly used solar thermal systems due to their simplicity, durability, and relatively low cost. These systems convert solar radiation into thermal energy by heating a working fluid that circulates through a network of tubes beneath an absorber plate. A transparent cover reduces heat loss to the environment while insulation minimizes energy loss from the back and sides [1]. Despite their widespread use, conventional flat plate solar collectors suffer from several limitations, including significant heat loss, limited efficiency during periods of low solar radiation, and inconsistent output temperatures. These challenges restrict their performance and make them less reliable in regions with highly variable solar intensity [2].

To overcome these limitations, this project proposes the design and development of an optimized flat plate solar collector system that integrates advanced materials and technologies to enhance thermal performance and energy storage capacity. The system incorporates copper pipes for superior thermal conductivity, a tank for lightweight and corrosion-resistant storage and a solar module to power the pump when light is available thus heat can be absorbed by the water via the flat solar collector. Which can be easily visualized in Figure 1.



Figure 1. *The actual project designed.*

2. Objectives

The objective of this project is to design and develop a flat plate solar collector (FPSC) with an integrated phase change material (PCM) storage tank to enhance thermal energy storage and improve the efficiency of solar water heating systems. While also being self-sustainable due to the pump being powered via a solar module which increases its reliability. The system is designed to address limitations in conventional flat plate collectors by incorporating latent heat storage, which allows for improved heat retention and a more consistent thermal output [3].

The specific objectives of this project include:

- **Achieving a Target Outlet Temperature of 60–80 °C:** The collector will be designed to efficiently absorb and transfer solar energy to the working fluid, ensuring a reliable and consistent thermal output suitable for various applications such as domestic water heating and industrial processes.
- **Optimizing Heat Absorption and Transfer:** The system will utilize an aluminum absorber plate and copper tubing in a serpentine flow configuration to enhance thermal conductivity and maximize heat transfer efficiency (Figure 2).

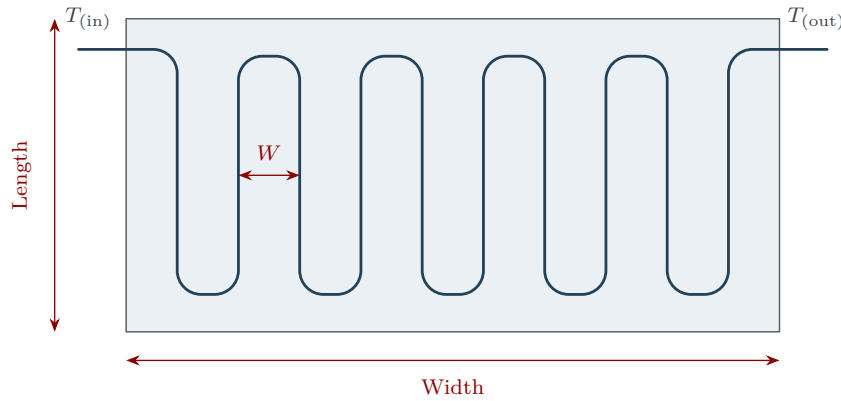


Figure 2. Schematic diagram of flat plate serpentine design.

- **Integrating a PCM Storage Tank for Thermal Energy Storage:** The incorporation of a PCM tank will enable the system to store excess heat during peak solar hours and release it when solar radiation is insufficient, thereby improving energy utilization and reducing reliance on auxiliary heating sources (Figure 3).

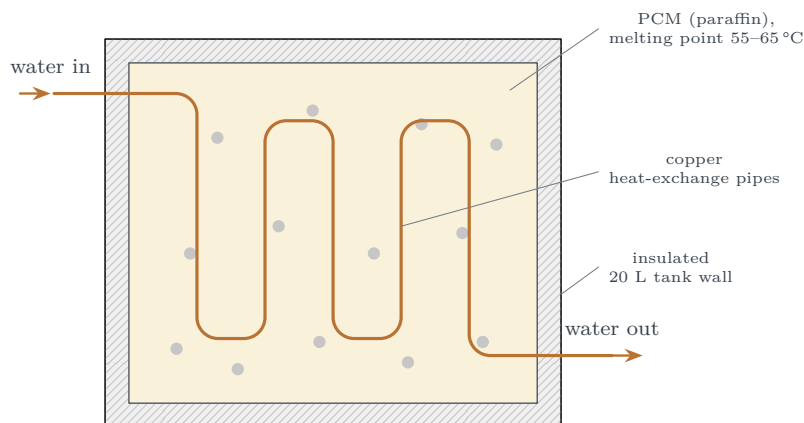


Figure 3. PCM storage tank.

- **Minimizing Heat Losses and Enhancing System Efficiency:** The design will focus on optimizing insulation, selecting appropriate materials, and incorporating an efficient heat exchange mechanism to minimize losses and enhance the overall efficiency of the collector.
- **Evaluating the Thermal Performance of the System:** The project will include an experimental analysis to assess the efficiency, heat retention capabilities, and performance of the system under varying solar conditions. The results will be compared to conventional FPSCs to determine the effectiveness of the PCM integration.
- **Promoting Sustainable and Renewable Energy Solutions:** By improving the performance and storage capabilities of solar collectors, the project aims to contribute to the adoption of renewable energy technologies and support the transition toward more sustainable energy systems.

3. Literature Review

Solar thermal energy has been extensively researched as a sustainable and environmentally friendly energy source. Flat plate solar collectors (FPCs) are among the oldest and most widely used solar thermal systems due to their relatively simple design and affordability. However, despite their long history of use, there are still significant opportunities to improve their performance through material innovation, advanced heat transfer fluids, and thermal storage enhancements [4].

3.1 Flat Plate Solar Collector Design

A typical flat plate solar collector consists of an absorber plate, fluid-carrying tubes, a transparent cover, and insulation. The absorber plate is usually coated with a selective surface to increase its ability to absorb solar radiation while minimizing heat losses through infrared radiation. The transparent cover reduces convective heat losses while allowing most of the incoming solar radiation to pass through. Insulation at the back and sides of the collector minimizes conductive heat losses [5].

Research has shown that tube configuration plays a crucial role in determining the overall efficiency of the collector. Serpentine layouts offer improved thermal performance compared to parallel tube configurations due to their ability to maintain a more uniform temperature distribution across the collector surface [6]. The main components of the collector are shown in Figure 4.

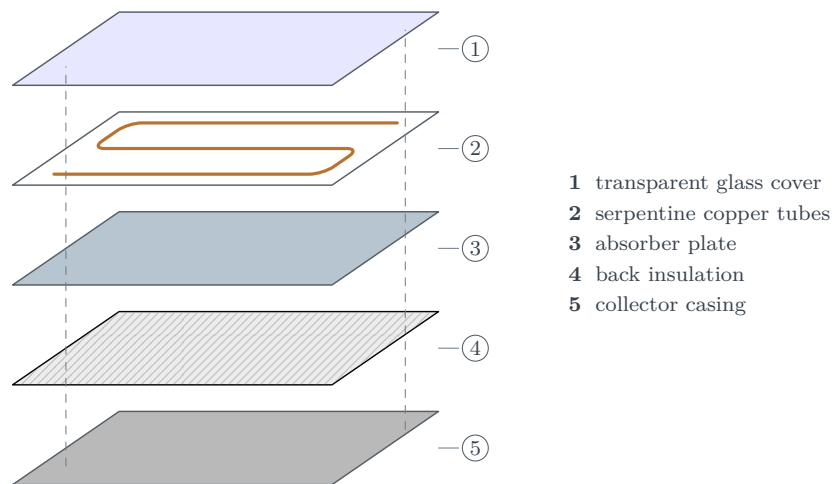


Figure 4. Flat plate collector components.

3.2 Phase Change Materials (PCMs)

One of the most significant advancements in solar collector design is the integration of phase change materials (PCMs) for thermal energy storage. PCMs have the unique ability to store large amounts of energy during phase transitions (from solid to liquid and vice versa), allowing the system to maintain a stable output temperature. The melting point of the PCM is carefully selected to match the desired operating temperature range of the collector [7].

In this project, a PCM with a melting point of 65 °C is used to ensure that the stored heat can be released when the working fluid temperature drops below the desired level. Studies have demonstrated that PCM-enhanced collectors can improve thermal storage capacity by 30–50%, making them highly effective for applications requiring consistent thermal output [8]. The working principle is illustrated in Figure 5.

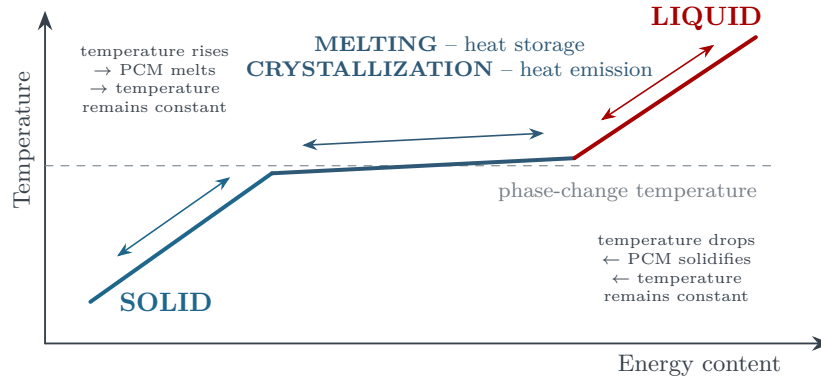


Figure 5. Working of phase change materials.

3.3 Solar Module

We have used a solar module as the choice to power this project due to it providing clean energy for our pump. The EURONET EU-M20W [9] is a 20 W monocrystalline solar module designed for small-scale solar applications. There are many types of solar modules but the more commonly used are two, monocrystalline and polycrystalline; the one used in this project is of the monocrystalline type due to them having generally higher efficiency in most small-scale applications. We will explain the differences between them now [10].

3.3.1 Monocrystalline Modules

Monocrystalline modules are made from single-crystal silicon, offering higher efficiency (17–22.8%) and better performance in high temperatures. They are more expensive due to a complex manufacturing process but provide longer lifespans and better output in limited space. Ideal for high-performance, space-constrained installations [11]. The module used in this project is shown in Figure 6.



Figure 6. *The monocrystalline module used in this project.*

3.3.2 Polycrystalline Modules

Polycrystalline modules are made from melted silicon fragments, resulting in lower efficiency (13–16%) and less performance in heat. They are more affordable and easier to produce, making them suitable for large-scale projects with fewer space restrictions and budget constraints [12]. Both cell technologies are compared in Figure 7.

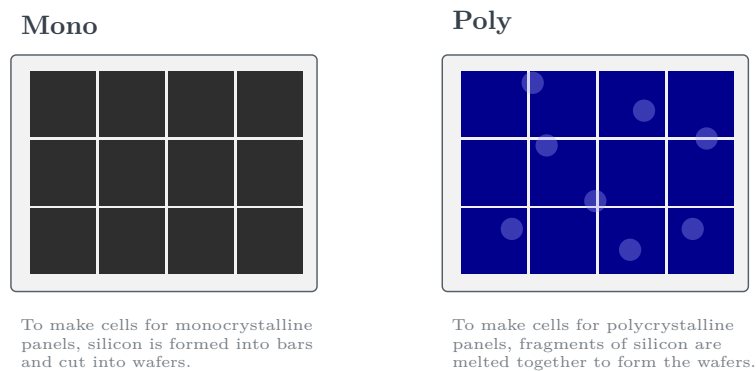


Figure 7. *Monocrystalline and polycrystalline solar modules.*

3.4 DC–DC Converters

3.4.1 Buck Converters

A buck converter is a DC–DC power converter that steps down voltage while increasing current. It operates using a switching element (typically a transistor), a diode, an inductor, and a capacitor to regulate output voltage efficiently. This type of converter is especially useful in low-voltage applications such as microprocessor power supplies and battery-operated devices. Its simplicity, high efficiency, and fast response make it ideal for modern electronic systems.

3.4.2 Boost Converters

A boost converter is used to step up a low input DC voltage to a higher output voltage. It stores energy in an inductor during the switch-on phase and releases it when the switch is off, thereby increasing voltage. Boost converters are widely used in systems like battery-powered devices, electric vehicles, and renewable energy setups where voltage needs to be elevated.

3.4.3 Buck–Boost Converters

The buck–boost converter combines the principles of both buck and boost converters, allowing it to either increase or decrease the input voltage depending on circuit demands. This makes it particularly suitable for applications where the input voltage can vary significantly around the required output level, such as in solar PV systems and battery-powered electronics. The converter uses a single switch, a diode, an inductor, and a capacitor, and operates in either continuous or discontinuous mode depending on the load. In this project we have used the buck–boost converter shown in Figure 8 due to its ability to combine both converters to ensure that water circulates throughout the day even when insufficient radiation is slightly affecting the solar module’s performance [13].

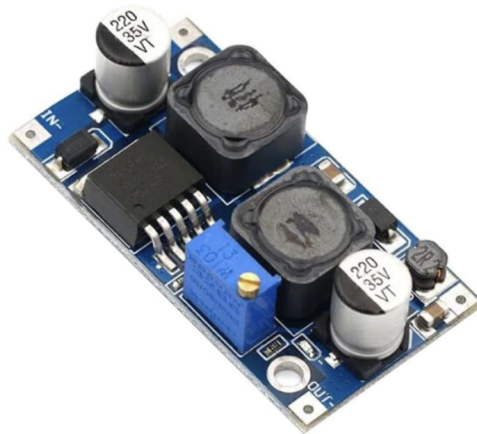


Figure 8. Shows the buck–boost converter used in this project.

3.4.4 Temperature Switch

Temperature switches are devices designed to monitor and control temperature by opening or closing an electrical circuit at a preset temperature threshold. They are essential in preventing overheating or overcooling in various applications [14]. The primary types include:

1. **Bimetallic Switches:** Utilize two metals with different thermal expansion coefficients; as temperature changes, the metals bend to actuate the switch.
2. **Gas-Actuated Switches:** Employ a sealed gas within a bulb or capsule that expands or contracts with temperature variations, triggering the switch mechanism.
3. **Electronic Switches:** Incorporate sensors like thermistors or RTDs, coupled with electronic circuits, to detect temperature changes and control the switch [15].

We will be using the last type, an electronic switch with a thermal sensor to detect the temperature of the water in the tank and circulate it in the solar collector when our desired temperature drops (Figure 9).

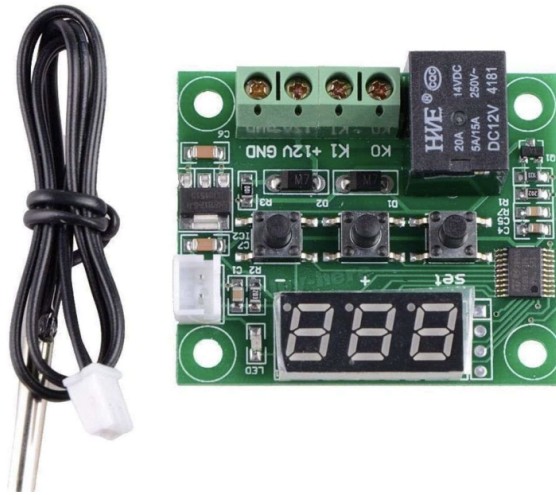


Figure 9. *Temperature switch used in this project.*

3.4.5 DC Brushless Water Pump

Brushless DC (BLDC) water pumps employ permanent magnet synchronous motors that eliminate the need for mechanical brushes, resulting in reduced mechanical wear and enhanced reliability. These pumps are particularly advantageous in applications requiring high efficiency which is very important in PV solar applications reaching 90% pump efficiency and the low maintenance required is very helpful, such as solar-powered water pumping systems. The integration of BLDC motors with photovoltaic arrays allows for efficient energy conversion and consistent water delivery, even under varying solar irradiance conditions. Advanced control techniques, including maximum power point tracking (MPPT) and sensorless operation, further optimize performance and energy utilization. In Figure 10 we show the PMSM DC brushless pump that has been used in this project that requires 12 V and 4.2 W which will be delivered by our solar module [16].



Figure 10. *DC pump used in this project.*

4. Strategy and Plan of Action

The successful implementation of the flat plate solar collector project requires a structured and well-defined strategy. This strategy is broken into multiple phases, from initial design and simulation to experimental testing and final optimization. Each phase builds on the previous one to ensure that the system meets performance expectations while minimizing thermal losses and maximizing efficiency.

The following four-phase approach is adopted:

4.1 Phase 1: Design and Component Selection

The first phase focuses on the selection of materials, key components, and the overall system configuration. Careful consideration is given to ensure optimal performance and durability.

- **Material Selection:** Copper pipes, aluminum storage tanks and PCM are chosen based on their high thermal conductivity, stability, and ability to improve heat transfer.
- **System Design:** A serpentine tube configuration is selected for uniform heat transfer across the absorber plate. The aluminum tank is designed to house PCM with a melting point of 65 °C for thermal buffering.
- **Insulation:** High-performance insulation materials are selected to minimize conductive and convective heat losses from the system.

4.2 Phase 2: Computational Analysis (CFD Simulations)

This phase involves the use of Computational Fluid Dynamics (CFD) simulations to analyze and optimize the performance of the flat plate solar collector under different operating conditions.

- **Fluid Flow and Heat Transfer Simulation:** CFD simulations are conducted to assess temperature distribution, pressure drop, and entropy generation within the system. These simulations provide insights into performance bottlenecks and areas of high energy loss [17].
- **Optimization of Design Parameters:** Critical parameters such as pipe diameter, mass flow rate, and PCM configuration are adjusted to maximize thermal efficiency [18].
- **Boundary Conditions:** Inlet temperature is set at 25 °C, with solar radiation intensity applied at 800 W/m². Mass flow rates are varied to identify the optimal operating condition.

4.3 Phase 3: Prototype Development and Experimental Testing

The third phase involves building a physical prototype of the flat plate solar collector and conducting experimental tests to validate the simulation results.

- **Prototype Fabrication:** The prototype is constructed using the selected materials and optimized design from Phase 1 and Phase 2. Care is taken to ensure accuracy in dimensions and material application.
- **Testing Conditions:** The system is tested under controlled conditions and outdoor environments to evaluate performance metrics such as inlet/outlet temperatures, pressure drop, solar radiation intensity, and mass flow rate.
- **Data Collection and Analysis:** Experimental data is collected and analyzed to compare it with CFD predictions. Any discrepancies are addressed by fine-tuning the system design.

Deliverables for this phase include performance data, analysis reports, and system adjustments.

4.4 Phase 4: Performance Optimization and Final Validation

The final phase focuses on improving system performance based on the results of the experimental tests.

- **Optimization Adjustments:** The mass flow rate, insulation and thickness concentration are adjusted to enhance efficiency.
- **Long-Term Durability Tests:** The system is tested for long-term stability and durability under varying environmental conditions to ensure reliability.
- **Scalability Assessment:** The design is evaluated for scalability and adaptability to different applications, ranging from residential water heating to large-scale industrial processes.

5. Risk Assessment and Mitigation Strategies

The successful implementation of the flat plate solar collector system depends on identifying potential risks early and addressing them with effective mitigation strategies. Below are the key risks associated with the project and the measures taken to minimize them:

5.1 Material Failure

Risk: Materials used in the construction of the system (such as copper pipes, aluminum tanks, and insulation) may fail due to thermal fatigue, corrosion, or mechanical stress. This can compromise the structural integrity and thermal performance of the system.

Mitigation Strategies:

- **High-Quality Material Selection:** Ensure that copper pipes are of high-grade quality with corrosion-resistant coatings to prevent degradation.
- **Thermal Stress Testing:** Simulate and test thermal cycling conditions to verify the durability of the materials.
- **Regular Inspection and Maintenance:** Implement a regular inspection schedule to detect early signs of wear and tear.

5.2 Inaccurate Computational Fluid Dynamics (CFD) Results

Risk: CFD simulations may not fully capture the real-world behavior of the system due to assumptions and simplifications, leading to inaccurate predictions.

Mitigation Strategies:

- **Experimental Validation:** Build and test a prototype under controlled and real-world conditions to validate and correct the CFD predictions.
- **Iterative Refinement:** Use experimental data to fine-tune the CFD model and improve its accuracy.
- **Sensitivity Analysis:** Perform sensitivity analysis to understand how different assumptions affect the results and reduce uncertainty.

5.3 Heat Loss

Risk: Heat loss through the absorber plate, storage tank, and fluid-carrying pipes can reduce the overall efficiency of the system.

Mitigation Strategies:

- **High-Performance Insulation:** Use advanced insulation materials (e.g., polyurethane foam, fiberglass) with low thermal conductivity to minimize conductive heat loss.

- **Optimized Flow Geometry:** Design the pipe network and flow channels to reduce convective and radiative losses.
- **Thermal Imaging:** Conduct thermal imaging tests to identify and fix potential heat loss points.

5.4 Pressure Drop and Flow Instabilities

Risk: Excessive pressure drop across the pipe network can reduce fluid flow rate, leading to uneven heat distribution and decreased performance.

Mitigation Strategies:

- **Flow Rate Optimization:** Optimize the pipe diameter and layout to ensure a balance between heat transfer efficiency and pressure drop.
- **Mass Flow Monitoring:** Implement sensors to monitor mass flow rate and detect flow irregularities.
- **CFD Analysis:** Use CFD simulations to predict and mitigate pressure drop in the design phase.

5.5 PCM Performance Degradation

Risk: The phase change material (PCM) may degrade over time due to repeated melting and solidification cycles, reducing its thermal storage capacity.

Mitigation Strategies:

- **Use High-Quality PCMs:** Select PCMs with high thermal stability and resistance to degradation.
- **Testing for Longevity:** Conduct long-term performance tests to assess the life cycle of the PCM.
- **Modular Tank Design:** Design the storage tank with replaceable PCM modules for easier maintenance and upgrades.

5.6 Data Acquisition and Sensor Failure

Risk: Inaccurate or incomplete data collection due to sensor failure or calibration errors can affect system monitoring and performance analysis.

Mitigation Strategies:

- **Redundant Sensors:** Use multiple sensors for critical parameters such as temperature, flow rate, and pressure to improve reliability.
- **Regular Calibration:** Establish a regular calibration schedule to maintain sensor accuracy.
- **Automated Monitoring System:** Integrate an IoT-based monitoring system for real-time data collection and alerts for sensor malfunctions.

5.7 Environmental and Weather Conditions

Risk: Unpredictable weather conditions such as prolonged cloud cover or heavy rain can affect the system's thermal performance.

Mitigation Strategies:

- **PCM-Based Storage:** Use PCM to store thermal energy during periods of high solar radiation, ensuring stable output even during cloudy periods.
- **Weather Data Integration:** Use real-time weather data to adjust system parameters and optimize performance.

- **Backup Heating System:** Consider integrating a backup heating system to ensure continuous operation during extended periods of low solar radiation.

5.8 Cost and Manufacturing Constraints

As university students we did the best we could provide by our local market and our financial abilities.

- **Cost constraints:** We have found that the higher the material grade was the purer and more efficient it is and thus we faced financial constraints.
- **Manufacturing constraints:** The solar collector and PCM storage tank used in this project were designed by us but manufactured by local shops and we faced constraints due to their ability to have the tools needed and the experience to manufacture the parts in a professional manner, although some could but that would lead us to our first constraint, the cost.

6. Key Milestones and Timeline

The following timeline outlines the key milestones for the flat plate solar collector project. The tasks are distributed across the remaining weeks to ensure the project is completed on time while allowing for iteration and improvements.

Week 8 (Feb 15–21, 2025): Finalize Design and CFD Simulation Setup

- Finalize the detailed design of the flat plate solar collector.
- Configure and validate the Computational Fluid Dynamics (CFD) simulation environment.
- Define boundary conditions and initial parameters for the simulation (e.g., mass flow rate, solar radiation, inlet temperature).
- Begin preliminary simulations to assess temperature distribution and fluid flow patterns.

Weeks 9–10 (Feb 22 – Mar 7, 2025): Complete CFD Simulations and Analyze Results

- Run multiple simulation scenarios with varying design parameters (e.g., pipe diameter, PCM configuration, and mass flow rate).
- Analyze CFD results to identify performance bottlenecks and optimize the design.
- Focus on minimizing entropy generation and pressure drop while maximizing thermal efficiency.

Deliverables: CFD Report, Optimized Design Parameters.

Week 11 (Mar 8–14, 2025): Build Prototype and Prepare for Testing

- Fabricate the prototype using the selected materials (copper pipes, aluminum tank, PCM, nanofluids).
- Ensure proper insulation and accurate installation of sensors (temperature, pressure, and mass flow rate).
- Conduct preliminary checks for leaks and material integrity.

Weeks 12–13 (Mar 15–28, 2025): Conduct Experimental Testing

- Test the prototype under controlled and outdoor conditions.
- Measure performance parameters such as inlet/outlet temperatures, solar radiation intensity, and mass flow rate.

- Collect data to validate the CFD results and identify discrepancies.

Deliverables: Experimental Data Report, Performance Comparison with CFD Predictions.

Week 14 (Mar 29 – Apr 4, 2025): Analyze Data and Optimize the System

- Analyze experimental data to determine the overall efficiency and performance of the system.
- Make adjustments to system components based on the test results (e.g., modify flow rate or improve insulation).
- Finalize the optimized design for documentation.

Week 15 (Apr 5–11, 2025): Draft Final Report and Prepare Presentation

- Begin writing the final report, incorporating design methodology, CFD analysis, experimental results, and conclusions.
- Prepare visual aids, graphs, and images for the report.
- Draft the presentation slides for the final project defense.

Week 16 (Apr 12–28, 2025): Final Report Submission and Presentation

- Review and edit the final report for accuracy and completeness.
- Submit the final report on April 28, 2025.
- Practice and deliver the final presentation.

7. Design Methodology and Equations

The design methodology for the proposed flat plate solar collector focuses on integrating high-performance materials, optimizing thermal efficiency through Computational Fluid Dynamics (CFD) simulations, and validating performance through experimental testing. The design process is divided into several key stages: material selection, thermal calculations, CFD analysis, and prototype testing [19].

7.1 Material Selection and Justification

7.1.1 Copper Pipes

Copper is selected for the fluid-carrying tubes due to its high thermal conductivity ($\sim 401 \text{ W/m}\cdot\text{K}$), which ensures rapid heat transfer from the absorber plate to the working fluid. This high conductivity significantly reduces thermal resistance, resulting in improved system efficiency. Additionally, copper's resistance to corrosion enhances the longevity of the system, especially in environments with varying humidity and temperature. The mechanical properties of copper—such as high tensile strength and flexibility—make it ideal for the serpentine tube configuration used in the collector. The serpentine layout ensures even heat distribution across the absorber plate, preventing temperature hotspots and improving overall heat absorption [20].

7.1.2 Water Tank

The thermal storage system consists of a 20-liter tank designed to house phase change material (PCM) for efficient thermal energy storage. Inside the tank, two copper pipes run through the volume of PCM, facilitating effective heat exchange between the PCM and the working fluid. Copper is chosen for its high thermal conductivity, which enables rapid heat transfer during both charging and discharging cycles. To ensure energy is retained for extended periods.

7.1.3 Phase Change Materials (PCMs)

PCMs are integrated into the tank to enhance thermal energy storage capacity. The PCM used in this project has a melting point of $55\text{--}65^\circ\text{C}$, which is ideal for domestic water heating applications. During periods of excess solar radiation, the PCM absorbs and stores heat as it melts. When the solar intensity decreases, the PCM solidifies and releases the stored energy, maintaining a stable output temperature for the working fluid. This thermal buffering effect ensures consistent performance even during cloudy periods or at night [6]. The paraffin used is shown in Figure 11.



Figure 11. *Paraffin PCM used in this project.*

7.2 Thermal Efficiency and Heat Transfer Calculations

The performance of the flat plate solar collector is evaluated using heat transfer and thermal efficiency equations. The key parameters include mass flow rate, useful heat gain, and thermal efficiency.

Equation and Calculation:

1. Heat Transfer and Useful Heat Gain Calculation The amount of useful heat gain from the flat plate solar collector is calculated using the following equation [22]:

$$Q_u = \dot{m} c_p (T_{out} - T_{in}) \quad (1)$$

where:

- Q_u = useful heat gain (W)
- $\dot{m}$ = mass flow rate (kg/s)
- c_p = specific heat capacity of the fluid (J/kg·K)
- T_{out} and T_{in} = outlet and inlet temperatures (°C)

2. The thermal efficiency is given by:

$$\eta = \frac{Q_{u, \text{useful}}}{Q_{\text{sun}, \text{in}}} \quad \text{or} \quad \eta = \frac{Q_u}{I A_c} \quad (2)$$

where:

- η = thermal efficiency
- I = solar radiation intensity (W/m²)
- A = collector area (m²)

3. Sensible Heat

$$Q_s = m \cdot C \cdot \Delta T \quad (3)$$

where:

- Q_s = sensible heat (J)
- m = mass of water (kg)
- c = specific heat of water (4186 J/kg·°C)

4. Latent Heat

$$Q_l = m \cdot L \quad (4)$$

where:

- Q_l = latent heat stored (J)
- m = mass of water (kg)
- L = latent heat of fusion (J/kg)

7.3 Experimental Tools

7.3.1 Solar Irradiance Meter

To accurately evaluate the performance of our solar water heating system, we utilized a solar irradiance meter shown in Figure 12 during the testing phase. This device measures the solar irradiance—the power per unit area received from the sun in the form of electromagnetic radiation on a given surface, typically expressed in watts per square meter (W/m^2). Monitoring solar irradiance was essential for correlating system performance with actual sunlight availability and ensuring consistency across test conditions.



Figure 12. *Solar irradiance meter.*

7.3.2 Thermometer

Temperature measurement was a fundamental part of our system testing and performance evaluation. For this purpose, we used a digital thermometer (Figure 13) equipped with thermocouple probes to monitor temperatures at various critical points in the system. This included the inlet and outlet of the flat plate solar collector and the temperature of the water storage tank.



Figure 13. *Digital thermometer.*

8. Testing and Results

8.1 Flat Plate

To evaluate the thermal performance of the designed flat plate solar collector, a full-day experimental test was conducted. The primary objective was to assess the collector's ability to increase the temperature of water flowing through it, and subsequently determine the useful energy gained and overall efficiency based on measured solar irradiance.

The flat plate solar collector was positioned outdoors with an optimal tilt to maximize solar exposure throughout the day. Water at a controlled inlet temperature of 23 °C was continuously circulated through the collector using a steady flow system. The mass flow rate was maintained at 0.06667 kg/s, and the specific heat capacity of water was taken as 4186 J/kg.°C.

Temperatures were recorded at both the inlet and outlet at half-hour intervals from 7:00 AM to 6:00 PM. Simultaneously, solar irradiance was measured to analyze performance over changing sunlight conditions.

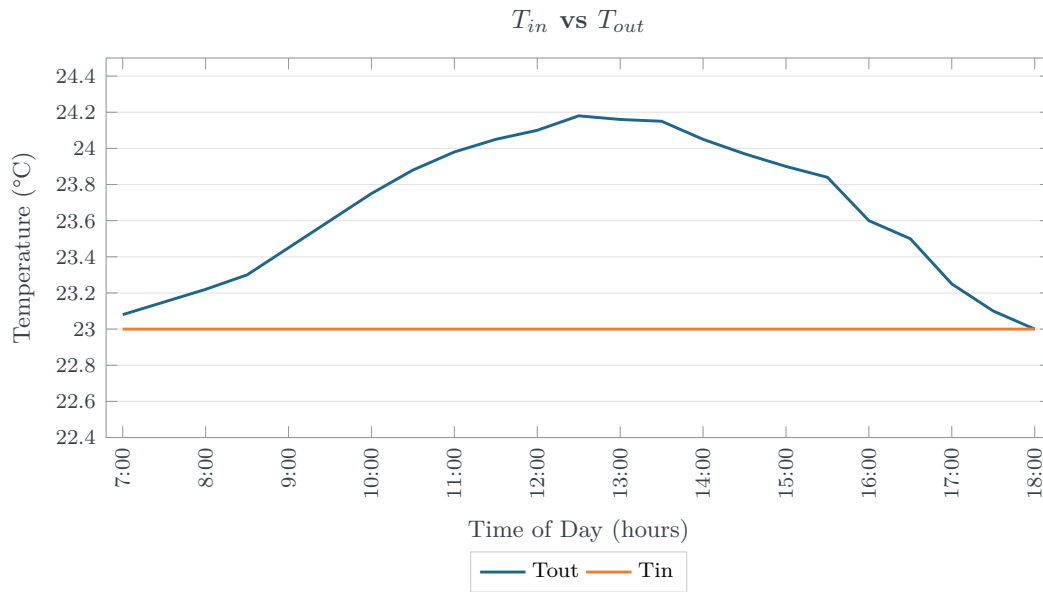


Figure 14. T_{in} vs T_{out} .

Figure 14 shows the variation of inlet and outlet water temperature throughout the day. The outlet temperature steadily increased as solar irradiance rose, peaking around noon, then gradually decreased as solar input declined in the afternoon.

Figure 15 compares the calculated useful heat gain Q_u with the theoretical heat input Q . The graph reflects strong alignment between available solar energy and thermal output, confirming that the collector's performance is directly dependent on solar irradiance levels.

Figure 16 illustrates the solar irradiance measured during the test. Peak irradiance occurred around 12:30 PM, correlating with the highest outlet water temperature and maximum useful heat gain.

The test results confirmed the flat plate collector's ability to effectively convert solar energy into thermal energy. The outlet water temperature increased consistently during periods of high irradiance, and the system demonstrated performance in line with expectations:

$$\eta = \frac{Q_u}{Q_s} = 63\% \quad (5)$$

where Q_u is the useful heat gained by the fluid and Q_s is the theoretical solar input energy. The overall efficiency of the system was found to be approximately 63%. This indicates that 63% of

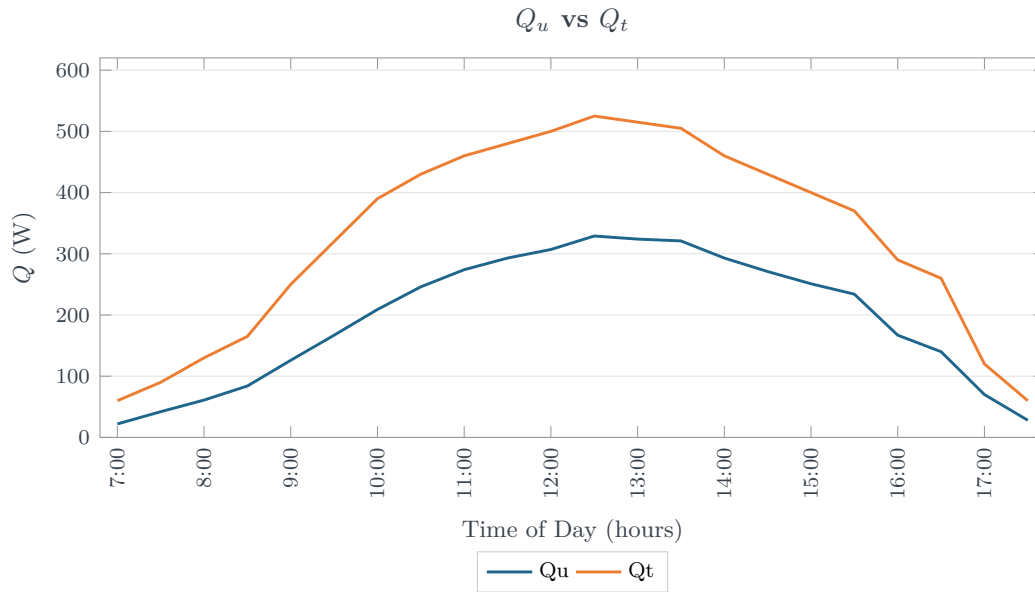


Figure 15. Q_u vs Q_t .

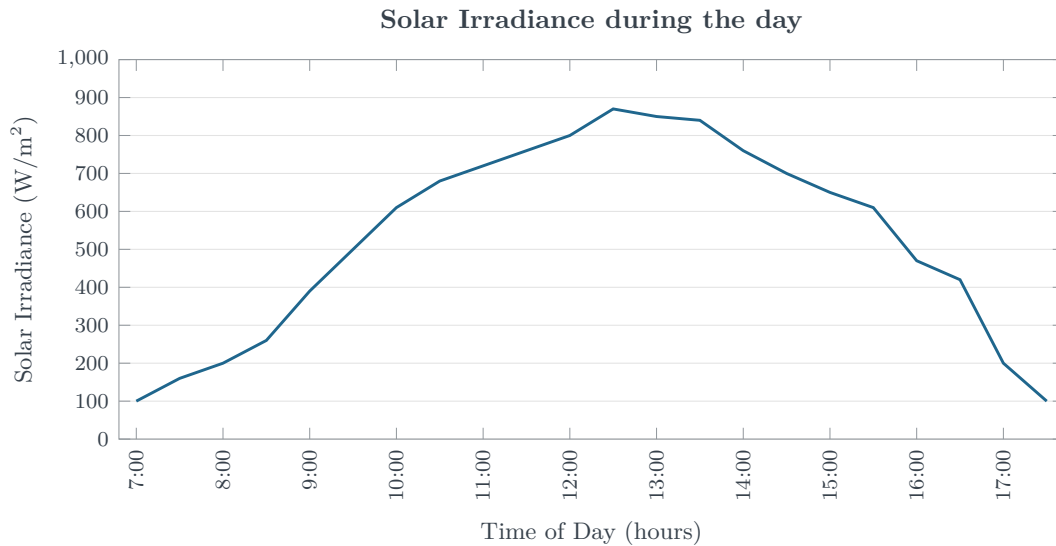


Figure 16. Solar irradiance.

the incident solar energy was successfully converted into usable thermal energy by the collector. The remaining energy was lost primarily due to thermal losses through conduction, convection, and radiation, which are typical in flat plate systems.

8.2 PCM Storage Tank

To evaluate the thermal storage performance of the Phase Change Material (PCM) storage system, a comparative test was conducted using two identical water tanks. The goal was to assess the effectiveness of integrating PCM—paraffin—in slowing down the rate of temperature drop during passive cooling. One tank (Tank A) was filled entirely with water, while the second tank (Tank B) included copper coils containing PCM.

Both tanks were initially at ambient room temperature (25 °C). At the start of the experiment, 80 °C water was added rapidly to both tanks to bring them to the same initial thermal condition. Temperature measurements were then recorded at half-hour intervals over a 12-hour period

using a thermometer placed at mid-height within each tank. Ambient conditions were monitored and remained consistent throughout the test. The data collected was used to generate the temperature–time curves and to calculate the total thermal energy retained in each tank using both sensible and latent heat equations.

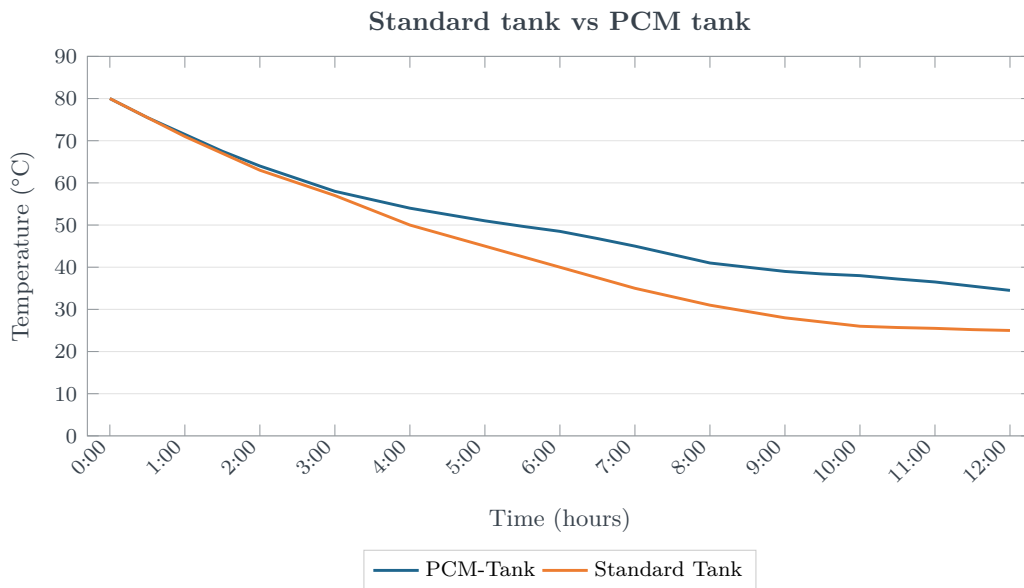


Figure 17. *PCM tank.*

Figure 17 shows the cooling curves of the two tanks over a 12-hour period, with Tank A (standard water tank) in orange and the PCM tank in blue.

After approximately 4 hours, the temperature of the PCM tank begins to deviate noticeably from that of the standard tank. The standard tank continues to cool steadily toward room temperature, while the PCM tank cools at a slower rate. This behavior is attributed to the presence of the phase change material, which absorbs a significant amount of thermal energy during its melting process without a substantial drop in temperature. This latent heat absorption delays the temperature decline of the PCM tank.

By the sixth hour, the PCM tank maintains a temperature around 49 °C, while the standard tank has already cooled down to approximately 28 °C. This difference widens over time, and by the end of the 12-hour period, the PCM tank stabilizes at approximately 36 °C, whereas the standard tank approaches the ambient temperature (around 25 °C).

These results demonstrate the effectiveness of incorporating phase change materials into thermal storage systems. The PCM significantly enhances the tank's ability to retain heat over longer periods, reducing the rate of heat loss and maintaining higher temperatures compared to a standard tank. This characteristic can be highly beneficial in applications where thermal energy needs to be stored and utilized over extended periods.

8.3 Overall System Performance

The complete solar water heating system, powered entirely by renewable energy, consists of a photovoltaic (PV) panel that supplies electricity to a DC converter, a temperature-controlled switch, and a circulation pump. During daytime operation, the PV panel generates sufficient power to drive the pump, circulating water through the flat-plate solar collector into the storage tank.

To automate system operation and ensure safe performance, a temperature sensor installed in the storage tank continuously monitors water temperature. When the tank water temperature reaches 80 °C, the control system automatically shuts off the circulation pump to prevent

overheating, protect the pump, and preserve system efficiency. Once the temperature falls below the setpoint, the system automatically restarts the pump when solar conditions allow, maintaining fully autonomous, user-free operation.

Based on experimental testing, the system is capable of heating approximately 30 liters of water per day, raising the temperature from an initial 23 °C up to 80 °C under typical sunny conditions with an average daily solar radiation of 6.2 kWh/m²/day. Integration of a Phase Change Material (PCM) inside the tank further enhances thermal retention, allowing hot water availability to extend several hours beyond sunset compared to conventional designs.

The system is fully sustainable, environmentally friendly, and provides domestic hot water entirely using renewable energy, by eliminating the need for grid electricity or fossil fuels.

9. Conclusion

This project successfully designed, developed, and experimentally validated a fully renewable solar water heating system that combines a flat-plate solar collector with a Phase Change Material (PCM)-enhanced thermal storage tank to improve energy capture and retention. The main objective was to achieve stable domestic water temperatures within the range of 60–80 °C while enhancing system efficiency and ensuring extended heat availability during periods of low or no solar radiation.

The flat-plate collector, constructed using high-conductivity copper piping beneath an aluminum absorber plate, demonstrated consistent thermal performance, achieving a measured thermal efficiency of approximately 63%. Experimental testing showed that the system could heat approximately 30 liters of water per day, raising the temperature from an initial 23 °C to 80 °C under typical sunny conditions with an average daily solar radiation of 6.2 kWh/m²/day.

Complementing the collector, the PCM storage tank utilized a paraffin-based phase change material to significantly enhance the thermal retention capabilities of the system. Comparative testing between the PCM-enhanced tank and a conventional water-only tank revealed that the PCM tank maintained a temperature nearly 9 °C higher after 12 hours of cooling, effectively extending the availability of hot water well beyond solar input hours.

Powered entirely by renewable solar energy and operating without any dependence on the electrical grid, the system provides a sustainable and environmentally friendly solution for domestic water heating. By reducing carbon emissions and promoting efficient use of renewable resources, this project directly supports the objectives of Saudi Vision 2030 in advancing clean energy technologies and sustainable development.



Figure 18. *Overall system.*

A. Appendix A: SDP Relevant Courses, Standards and Constraints

The relevant courses which are used for the project:

S.No.	Course codes and names
a.	REE 205 3D-CAD and Modeling
b.	REE 2051 3D-CAD and Modeling-Lab
c.	REE 304 Solar PV
d.	REE 306 Heat and Mass Transfer
e.	REE 401 Solar Thermal Energy
f.	REE ISE 465 Industrial Safety

References

- [1] Ramírez-Minguela, J.J., Rangel-Hernández, V.H., Alfaro-Ayala, J.A., Elizalde-Blancas, F., Ruiz-Camacho, B., López-Núñez, O.A., & Alvarado-Rodríguez, C.E. ‘Performance Comparison of Different Flat Plate Solar Collectors by Means of the Entropy Generation Rate Using Computational Fluid Dynamics.’ *Entropy*, 25(4), 621. <https://doi.org/10.3390/e25040621>
- [2] Alobaid, M., Hughes, B., Heyes, A., & O’Connor, D. ‘Determining the Effect of Inlet Flow Conditions on the Thermal Efficiency of a Flat Plate Solar Collector.’ Accessed: Feb. 21, 2025. [Online]. Available: <https://doi.org/10.3390/fluids3030067>
- [3] Elsheikh, A.H., Abd Elaziz, M., Shanmugam, B., Sathyamurthy, R., & Deepak, K. ‘Applications of Nanofluids in Solar Energy: A Review.’ *Renewable and Sustainable Energy Reviews*, 82, 3483–3502. <https://doi.org/10.1016/j.rser.2017.10.108>
- [4] Zhang, H., Liu, Y., Yang, Y., & Wang, X. ‘Thermal Performance Enhancement of a Flat Plate Collector Using PCM and Nanofluids.’ *Applied Thermal Engineering*, 103, 1285–1291. <https://doi.org/10.1016/j.applthermaleng.2016.05.178>
- [5] Aruna, A., & Rajendra Prasad, B. ‘Numerical Modeling of Heat Transfer in PCM-Enhanced Solar Collectors.’ *Energy Procedia*, 109, 276–283. <https://doi.org/10.1016/j.egypro.2017.03.082>
- [6] Noor, I., & Ishak, M. ‘Simulation and Experimental Validation of Thermal Energy Storage Using PCM.’ *Applied Thermal Engineering*, 111, 175–183. <https://doi.org/10.1016/j.applthermaleng.2016.09.097>
- [7] Natarajan, E., & Sathish, R. (2009). ‘Phase Change Material Selection for Thermal Energy Storage at Medium Temperature Applications.’ *Journal of Mechanical Science and Technology*, 23, 2006–2010. https://www.researchgate.net/publication/360643777_Phase_Change_Materials_Selection_for_Thermal_Energy_Storage_System_A_Review
- [8] Thakur, S., & Singh, M.K. (2013). ‘Optimization of Solar Energy Systems Using Computational Fluid Dynamics.’ *International Journal of Energy Research*, 37(13), 1501–1521. https://www.researchgate.net/publication/258221365_Dynamic_thermal_models_and_CFD_analysis_for_flat-plate_thermal_solar_collectors_-_A_review
- [9] China Solar KS-M20W Solar Panel Specifications. Accessed: Apr. 23, 2025. [Online]. Available: <https://energypal.com/best-solar-panels-for-homes/china-solar/ks-m20w>
- [10] O. Ayadi, R. Shadid, A. Bani-Abdullah, M. Alrbai, M. Abu-Mualla, and N. A. Balah, “Experimental comparison between Monocrystalline, Polycrystalline, and Thin-film solar systems under sunny climatic conditions,” *Energy Reports*, vol. 8, pp. 218–230, Nov. 2022, doi: 10.1016/J.EGYR.2022.06.121.
- [11] Monocrystalline vs Polycrystalline Solar Panels | American Solar Energy Society. Accessed: Apr. 23, 2025. [Online]. Available: <https://ases.org/monocrystalline-vs-polycrystalline-solar-panels/>
- [12] L. A. Dobrzański, M. Szczęsna, M. Szindler, and A. Drygała, “Electrical properties mono- and polycrystalline silicon solar cells”.
- [13] J. Almazán, N. Vázquez, C. Hernández, J. Alvarez, and J. Arau, “Comparison between the buck, boost and buck-boost inverters,” *International Power Electronics Congress – CIEP*, pp. 341–346, 2000, doi: 10.1109/CIEP.2000.891437.

- [14] What are the Different Types of Thermal Switches? Accessed: Apr. 23, 2025. [Online]. Available: <https://www.saftty.com/what-are-the-different-types-of-thermal-switches.html>
- [15] Types of Thermal Switches for Temperature Control. Accessed: Apr. 23, 2025. [Online]. Available: <https://www.thermal-engineering.org/3-types-of-thermal-switches-for-temperature-control/>
- [16] T. Rymarczyk et al., “Optimal Rotor Design and Analysis of Energy-Efficient Brushless DC Motor-Driven Centrifugal Monoset Pump for Agriculture Applications,” *Energies* 2024, Vol. 17, Page 2280, vol. 17, no. 10, p. 2280, May 2024, doi: 10.3390/EN17102280.
- [17] Zhang, R., & Tian, W. (2015). ‘Numerical Study of Heat Transfer in a Flat Plate Solar Collector with PCM.’ *Applied Energy*, 140, 1–9. https://www.researchgate.net/publication/258221365_Dynamic_thermal_models_and_CFD_analysis_for_flat-plate_thermal_solar_collectors_-_A_review
- [18] Mahmud, K., & Fraser, R.A. (2014). ‘Heat Transfer Enhancement in Flat Plate Collectors with Nanoparticle Suspensions.’ *Procedia Engineering*, 90, 686–691. https://www.researchgate.net/publication/258221365_Dynamic_thermal_models_and_CFD_analysis_for_flat-plate_thermal_solar_collectors_-_A_review
- [19] Wu, S.Y., Zhu, D.S., & Zhang, Q.L. (2010). ‘Dynamic thermal models and CFD analysis for flat-plate thermal solar collectors – A review.’ *Renewable and Sustainable Energy Reviews*, 14(1), 343–349. https://www.researchgate.net/publication/258221365_Dynamic_thermal_models_and_CFD_analysis_for_flat-plate_thermal_solar_collectors_-_A_review
- [20] Kreith, F., & Goswami, D. Y. (2016). *Principles of Solar Engineering* (3rd ed.).
- [21] Jayaraj, S., & Deepak, M. (2013). ‘Comparative Study of Thermal Behavior in PCM-Based Solar Collectors.’ *Energy Conversion and Management*, 73, 10–16. https://www.researchgate.net/publication/222883121_Solar_water_heaters_with_phase_change_material_thermal_energy_storage_medium_A_review
- [22] Duffie, J. A., & Beckman, W. A. (2013). *Solar Engineering of Thermal Processes* (4th ed.). John Wiley & Sons.